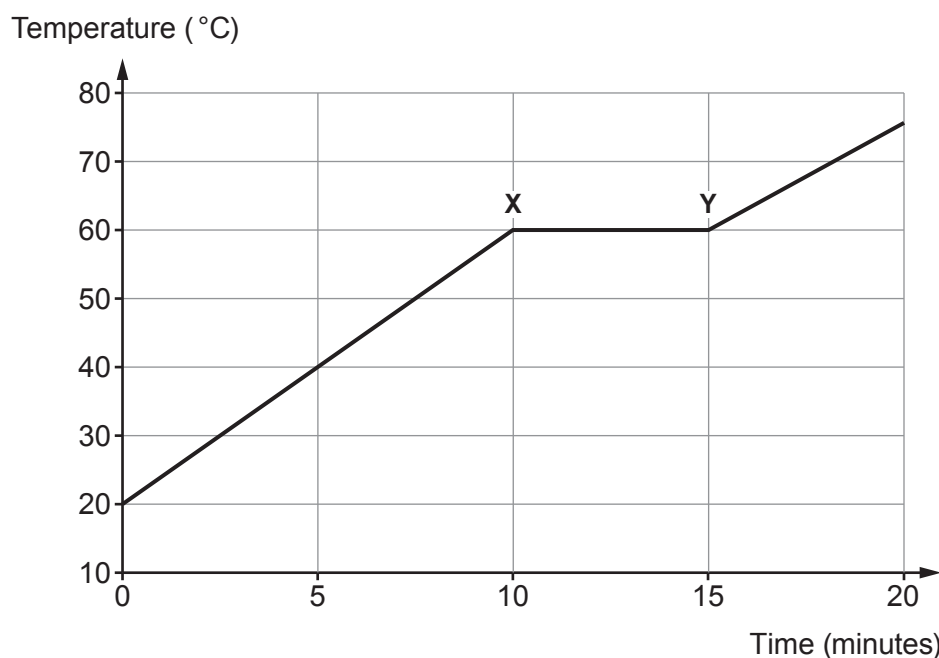


6. The graph below shows the results obtained when **36 g of solid wax** was heated using a heater. The temperature of the wax was recorded every minute.



- (a) Explain why the energy supplied from the heater between **points X and Y** doesn't produce a rise in temperature of the wax. [2]

.....

.....

.....

- (b) (i) Use the equation:

$$\text{thermal energy supplied} = \text{mass} \times \text{specific latent heat}$$

to calculate the amount of thermal energy supplied to the 36 g of wax between **points X and Y**. [2]

(Specific latent heat of fusion (L_f) for wax = 220 J/g)

Thermal energy supplied = J



- (ii) All the energy is transferred from the heater to the wax during the 5 minute interval between **X and Y**.

- I. Calculate the time, in seconds, of the interval between **X and Y**. [1]

Time = s

- II. Use the equation:

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

- to calculate the power of the heater. [2]

Power of heater = W

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